

PFL-AI Formula Sheet

PCI AI Project Finance Leader — the complete quantitative reference

NPV · IRR · DSCR · LLCR · PLCR · WACC

How to use this sheet

Every formula the PFL-AI Body of Knowledge teaches, in one place, with the domain where it is worked in full.

Three columns matter. **Formula** is the expression. **Meaning** is what it answers. **Watch for** is the mistake that formula attracts in practice.

This sheet is a study aid. The examination specification, including whether any formula reference is provided, is confirmed by the Institute for each published examination form.

Conventions. Cash flows are period-end unless stated. Rates are per period and must match the period of the cash flows — an annual rate applied to semi-annual flows is the most common error in project finance modelling. Ratios to 2 dp. Currency in USD unless stated.

The consistency rule. Discount rate, cash-flow period and compounding frequency must agree. If debt service is semi-annual, the discount rate is the semi-annual rate, and n counts half-years. Most coverage-ratio errors are period-mismatch errors, not formula errors.

1. Notation

SYMBOL	MEANING	UNIT
CF_t	Cash flow in period t	currency
r, n	Discount rate per period, number of periods	ratio, count
DF	Discount factor	ratio
AF	Annuity factor	ratio
NPV, IRR	Net present value, internal rate of return	currency, ratio
$MIRR$	Modified internal rate of return	ratio
$CFADS$	Cash flow available for debt service	currency
DS	Debt service — interest plus scheduled principal	currency
$DSCR$	Debt service cover ratio	ratio
$LLCR, PLCR$	Loan life and project life cover ratios	ratio
$DSRA, MRA$	Debt service and maintenance reserve accounts	currency
D, E, V	Debt, equity, total capital ($V = D + E$)	currency
R_d, R_e	Cost of debt, cost of equity	ratio
k_d	Discount rate used for cover ratios (typically cost of debt)	ratio
t	Tax rate (in cost-of-capital contexts)	ratio
$WACC$	Weighted average cost of capital	ratio
$MOIC$	Multiple on invested capital	ratio
$EBITDA$	Earnings before interest, tax, depreciation, amortisation	currency

2. Time value of money • Domain 3

FORMULA	MEANING	WATCH FOR	DOMAIN
$FV = PV(x) \times (1 + r)^n$	Compounding	—	3.1
$PV(x) = FV \div (1 + r)^n$	Discounting	—	3.1
$DF = 1 \div (1 + r)^n$	Discount factor	—	3.1
$AF = [1 - (1 + r)^{-n}] \div r$	Annuity factor — PV of 1 per period for n periods	Applying to a growing stream	3.2
$PV(\text{perpetuity}) = CF \div r$	Level perpetuity	—	3.2
$PV(\text{growing perpetuity}) = CF_1 \div (r - g)$	Gordon growth	Valid only where $g < r$	3.2
$\text{Effective annual rate} = (1 + r/m)^m - 1$	Nominal to effective, m compounds per year	Comparing a nominal to an effective rate	3.3
$\text{Real rate} \approx [(1 + \text{nominal}) \div (1 + \text{inflation})] - 1$	Fisher relation	The additive approximation drifts at high inflation	3.4
$\text{Payment} = \text{Principal} \div AF$	Level annuity instalment	—	3.2

Annuity factor check. AF must equal the sum of the individual discount factors. At 8% over 5 periods, $AF = 3.9927$, and the five discount factors sum to 3.9927. If they disagree, the period convention is wrong.

3. Investment appraisal • Domain 4

FORMULA	MEANING	WATCH FOR	DOMAIN
$NPV = \sum CF_t \div (1 + r)^t$	Net present value	Including financing flows in an unlevered NPV	4.1
$IRR: NPV = 0$	Internal rate of return	Multiple IRRs when signs change more than once	4.1
$MIRR = (FV \text{ of inflows at reinvest rate} \div -PV \text{ of outflows at finance rate})^{(1/n)} - 1$	Modified IRR	Mixing the two rates	4.1
$PI = PV(\text{inflows}) \div \text{Initial investment}$	Profitability index	—	4.2
$PI = 1 + NPV \div \text{Initial investment}$	Equivalent form — use as a check	—	4.2
$\text{Payback} = \text{Years before recovery} + (\text{Unrecovered} \div \text{Cash flow in that year})$	Simple payback	Ignores everything after the payback point	4.2
$\text{Discounted payback}$	As above, on discounted flows	—	4.2
$\text{Equivalent annual cost} = NPV \div AF$	Comparing unequal-life assets	—	4.3
$\text{Switching value} = \Delta\% \text{ in a variable that drives NPV to zero}$	Sensitivity breakeven	—	4.4

IRR pathologies — the three to know. (1) **Multiple IRRs** where the cash-flow sign changes more than once; the equation has more than one root. (2) **No IRR** where the flows never cross zero. (3) **Scale and reinvestment blindness** — IRR implicitly assumes reinvestment at the IRR itself, which is why MIRR exists and why NPV governs where the two disagree on ranking.

4. Cost of capital and structure · Domain 9

FORMULA	MEANING	WATCH FOR	DOMAIN
$WACC = (E \div V) \times Re + (D \div V) \times Rd \times (1 - t)$	Weighted average cost of capital	Tax shield applied to equity	9.1
$Re = Rf + \beta \times (Rm - Rf)$	CAPM cost of equity	Equity beta used where asset beta is required	9.1
$Gearing = D \div (D + E)$	Leverage as a proportion of capital	Confused with debt-to-equity	9.2
$Debt : equity = D \div E$	Leverage as a ratio	60% gearing is 1.5 : 1, not 60 : 40 as a ratio	9.2
$\beta_{asset} = \beta_{equity} \div [1 + (1 - t) \times D \div E]$	Ungearing beta	—	9.1
$Interest\ cover = EBIT \div Interest$	Income-based cover	EBITDA variant is a different covenant	9.2
$Leverage = Net\ debt \div EBITDA$	Leverage covenant	Gross vs net debt definition	10.2

5. Cash flow definitions · Domains 6, 14

The waterfall, in order. Each level is paid only after the one above.

ORDER	LEVEL
1	Operating expenses
2	Tax
3	Senior debt interest
4	Senior debt scheduled principal
5	Reserve account top-ups — DSRA, MRA
6	Subordinated / mezzanine debt service
7	Distributions to equity, subject to the lock-up test

FORMULA	MEANING	WATCH FOR	DOMAIN
$CFADS = EBITDA - Tax\ paid - \Delta\ Working\ capital - Maintenance\ capex$	Cash available for debt service	Deducting interest — interest sits below CFADS	6.3
$DS = Interest + Scheduled\ principal$	Debt service	Excluding fees where the covenant includes them	10.2
$Cash\ available\ to\ equity = CFADS - DS - Reserve\ top-ups$	Distributable cash	—	6.4
$Lock-up: distributions\ blocked\ if\ DSCR < lock-up\ threshold$	Distribution test	Lock-up threshold differs from default threshold	10.3
$DSRA\ target = Next\ 6\ or\ 12\ months\ of\ debt\ service$	Reserve sizing	—	10.3

CFADS is pre-financing. Interest and principal are what CFADS is measured against. Deducting interest before computing CFADS double-counts it and inflates every cover ratio. This is the most consequential definitional error in project finance.

6. Coverage ratios · Domain 10 — the flagship

FORMULA	MEANING	DOMAIN
$DSCR = CFADS \div DS$	Cover in a single period	10.2
$LLCR = [PV(CFADS \text{ over remaining loan life, at } kd) + DSRA] \div \text{Debt outstanding}$	Cover over the life of the loan	10.2
$PLCR = [PV(CFADS \text{ over remaining project life, at } kd) + DSRA] \div \text{Debt outstanding}$	Cover over the life of the project	10.2
Minimum DSCR	The lowest DSCR in any period — the binding constraint	10.2
Average DSCR	Mean across periods — informative, never the covenant	10.2

How the three differ, and why all three exist.

RATIO	WINDOW	ANSWERS
DSCR	One period	Can the project service debt this period?
LLCR	Remaining loan term	Is there enough cash over the loan's life to repay it?
PLCR	Remaining project life	Is there headroom beyond loan maturity — the tail?

PLCR is normally greater than LLCR, because it discounts cash flows over a longer window against the same debt balance. The gap between them is the **tail** — the cash-generating life of the project after the debt matures. A thin tail is a credit concern even when LLCR looks comfortable, because it removes the lender's room to reschedule.

Debt sizing from a target ratio

FORMULA	MEANING	DOMAIN
$\text{Debt service capacity}_t = CFADS_t \div \text{Target DSCR}$	Maximum affordable service each period	10.1
$\text{Debt capacity} = \sum [\text{Capacity}_t \div (1 + kd)^t]$	Sculpted debt quantum	10.1
$\text{Sculpted repayment: principal}_t = \text{Capacity}_t - \text{Interest}_t$	Repayment shaped to cash flow	10.1

Sculpting versus annuity. A sculpted profile sets debt service as a constant multiple of CFADS, so DSCR is flat at the target in every period. An annuity profile sets a constant instalment, so DSCR varies with CFADS. Sculpting maximises debt capacity for a given minimum DSCR — which is why project finance uses it and corporate lending largely does not.

FORMULA	MEANING	DOMAIN
$\text{Average debt life} = \sum (\text{Principal}_t \times t) \div \sum \text{Principal}_t$	Weighted average life	10.1
$\text{Tail} = \text{Project life} - \text{Loan life}$	Headroom beyond maturity	10.2
$\text{Gearing achieved} = \text{Debt capacity} \div \text{Total project cost}$	Resulting leverage	10.1

7. Equity returns • Domains 6, 15

FORMULA	MEANING	WATCH FOR	DOMAIN
Equity IRR: NPV of equity cash flows = 0	Return to shareholders	Computed on project rather than equity flows	6.4
Project IRR	Return on total project cost, pre-financing	—	4.1
MOIC = Total distributions ÷ Total equity invested	Money multiple	Ignores timing entirely	15.2
Equity IRR > Project IRR when Project IRR > cost of debt	The leverage effect	Leverage magnifies losses symmetrically	9.2
Payback to equity	Period to recover equity injections	—	15.2

Two IRRs, two questions. Project IRR asks whether the asset is worth building. Equity IRR asks whether the deal is worth doing. A strong equity IRR on a weak project IRR is a statement about leverage, not about the asset.

8. Construction phase • Domain 14

FORMULA	MEANING	WATCH FOR	DOMAIN
Drawdown _t = Cost incurred _t × Debt proportion	Pro-rata debt draw	Equity-first structures draw differently	14.1
IDC = Σ Interest on drawn balance during construction	Interest during construction	Omitted from project cost	14.2
Total project cost = Construction + IDC + Fees + Working capital + Reserves	Funding requirement	IDC and fees excluded	14.2
Cost to complete = Total budget – Costs incurred	Remaining requirement	Not the same as remaining committed	14.3
Funding adequacy: Available funds ≥ Cost to complete	The lender's construction test	—	14.3
EAC (see PCL-AI sheet §4.2)	Forecast construction outturn	The controls and finance forecasts must reconcile	8.2

The bridge to project controls. A lender's cost-to-complete test and a project controls EAC answer the same question from two sides. If the controls function forecasts an outturn above the funded amount, the funding adequacy test has already failed — usually before anyone in the finance team has been told.

9. Risk and sensitivity · Domain 11

FORMULA	MEANING	DOMAIN
$EMV = Probability \times Impact$	Expected monetary value	11.2
$Switching\ value = \Delta\% \text{ to drive NPV or minimum DSCR to its limit}$	Breakeven sensitivity	11.3
$Sensitivity = \Delta\ Output \div \Delta\ Input$	Elasticity of the metric to a driver	11.3
$Downside\ case: \text{minimum DSCR} \geq 1.00$	Base survival test	11.3
$Breakeven\ tariff / volume$	Price or throughput at which DSCR hits the covenant	7.2

The standard sensitivity set. Capex overrun, delay to commercial operation, revenue or tariff reduction, opex increase, interest-rate movement, and availability or throughput shortfall — each run singly, then in a combined downside case.

10. Worked micro-examples

Each is internally consistent and reproducible from this sheet.

Appraisal. Cash flows $-1,000 \cdot 300 \cdot 400 \cdot 500 \cdot 400$, discount rate 10%.

- $NPV = 252.17$
- $IRR = 20.50\%$ (NPV at that rate is zero)
- $MIRR$ at 8% finance / 10% reinvestment = **16.36%**
- $PI = 1,252.17 \div 1,000 = 1.252 = 1 + 252.17 \div 1,000$ — the two forms agree
- $Payback = 2 + 300 \div 500 = 2.60\text{ years}$

Discount and annuity factors at 8% over 5 periods: $DF = 0.6806 \cdot AF = 3.9927$

DSCR. CFADS 42,000,000 · principal 18,000,000 · interest 9,500,000.

- $DS = 27,500,000 \rightarrow DSCR = 42,000,000 \div 27,500,000 = 1.53$

LLCR and PLCR. Debt outstanding 210,000,000 · DSRA 12,000,000 · kd 7% · CFADS 38,000,000 for 8 remaining loan years, then 36,000,000 for a further 7 project years.

- PV of CFADS over loan life = 226,909,000 → $LLCR = (226,909,000 + 12,000,000) \div 210,000,000 = 1.14$
- PV of CFADS over project life = 339,828,000 → $PLCR = (339,828,000 + 12,000,000) \div 210,000,000 = 1.68$
- The gap is the seven-year tail. $PLCR > LLCR$, as it should be.

WACC. E 400,000,000 · D 600,000,000 · Re 13.5% · Rd 7.0% · tax 25%.

- Gearing = $600 \div 1,000 = 60\%$ (debt : equity = 1.5 : 1)
- $WACC = 0.40 \times 13.5\% + 0.60 \times 7.0\% \times 0.75 = 5.40\% + 3.15\% = 8.55\%$

Debt sizing. Target DSCR 1.35 · kd 7% · CFADS 40.0, 42.0, 44.0, 43.0, 41.0 million over five years.

- Capacity each year = $CFADS \div 1.35 \rightarrow 29.63, 31.11, 32.59, 31.85, 30.37$ million
- Debt capacity = PV of that stream at 7% = **127,423,000**
- Check: servicing exactly that capacity returns a DSCR of **1.35 in every year** — the definition of a sculpted profile.

11. The ten formulas most often misapplied

1. **Interest deducted before CFADS.** CFADS is pre-financing. Deducting interest double-counts it and inflates every cover ratio.
2. **Period mismatch.** An annual rate applied to semi-annual debt service. Rate, period and compounding must agree.
3. **LLCR and PLCR treated as interchangeable.** They discount over different windows against the same debt. The gap is the tail, and the tail is a credit question.
4. **Average DSCR quoted as the covenant.** The **minimum** DSCR is the binding constraint. An average conceals the one period that breaches.
5. **IRR ranked above NPV on mutually exclusive projects.** IRR assumes reinvestment at itself and is blind to scale. NPV governs.
6. **Multiple IRRs unnoticed.** More than one sign change means more than one root. Check the sign pattern before quoting an IRR.
7. **Gearing confused with debt-to-equity.** 60% gearing is 1.5 : 1.
8. **Tax shield applied to the equity term of WACC.** $(1 - t)$ attaches to the cost of debt only.
9. **IDC and fees omitted from total project cost.** The funding requirement is larger than the construction contract, often materially.
10. **Equity IRR presented as project performance.** It is a statement about leverage as much as about the asset. Quote both.

12. Cross-references

FOR THE FULL TREATMENT	SEE
Every formula worked with figures and exercises	The PFL-AI Body of Knowledge, at the domain cited
Coverage ratios, sculpting and covenants in depth	PFL-AI Domain 10 — the quantitative flagship
The construction-phase bridge to project controls	PFL-AI Domain 8 and Domain 14
Cost forecasting that feeds the funding adequacy test	PCL-AI Formula Sheet, §4.2

Website — <https://projectcontrolsinstitute.org>

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